D4.2 Stability and change

D4 Continuity and change
Ecosystems

HL

Understandings:

- D4.2.12—Ecological succession and its causes
- D4.2.13—Changes occurring during primary succession
- D4.2.14—Cyclical succession in ecosystems
- D4.2.15—Climax communities and arrested succession

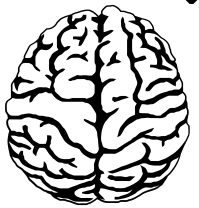
Key words:
Primary succession
Secondary succession
Climax community
Pioneer species
Cyclic succession
Arrested succession
Plagioclimax

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



D4.2.12—Ecological succession and its causes

What do all these pictures have in common?

















If left undisturbed, every environment will have a natural type of stable ecosystem (often forest)

Think of some causes of disruption and divide them into abiotic and biotic

Abiotic

Fire

Landslide

Volcano eruption

Glacier movement

Flood

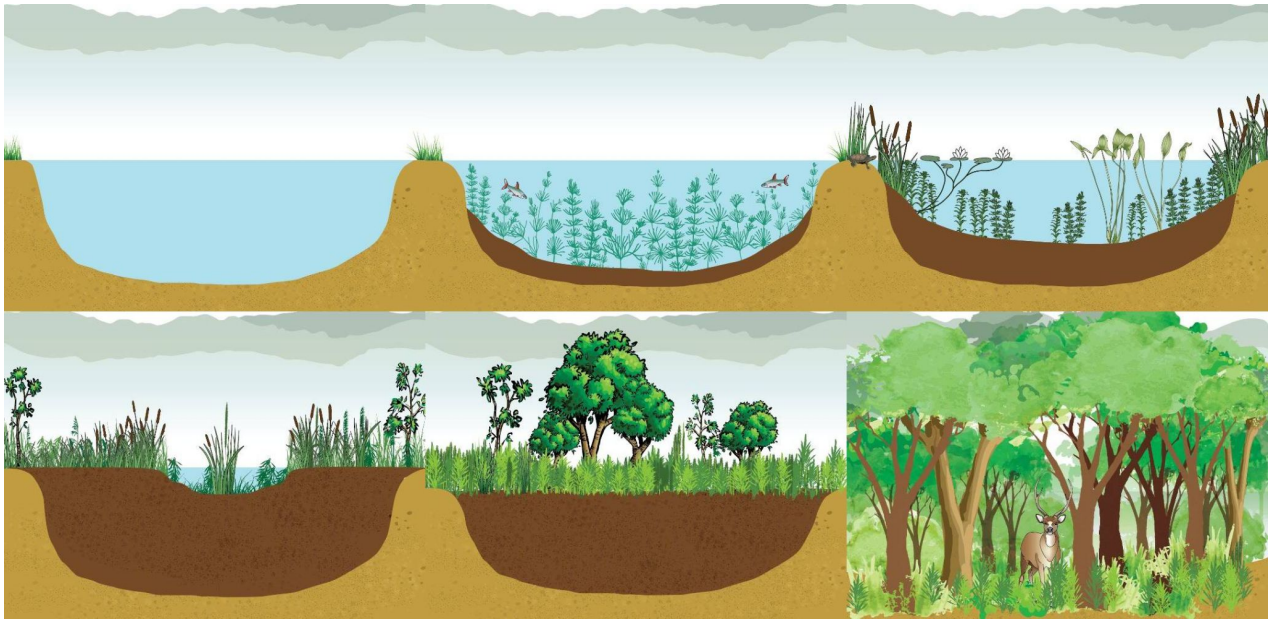
Tornado

Biotic

Human development

Overgrazing

When an environment is recovering from a change to its natural state it is said to be undergoing the process of **succession**



D4.2.13—Changes occurring during primary succession

Succession

“The progressive replacement of one dominant type of species or community by another in an ecosystem until a stable **climax community** is established.”

Primary succession

Development of a new biological community from a barren starting point with no soil

Secondary succession

Recolonisation of an environment after a dramatic disturbance. Soil present.



Can you think of some examples of causes for primary and secondary succession?

Primary succession:

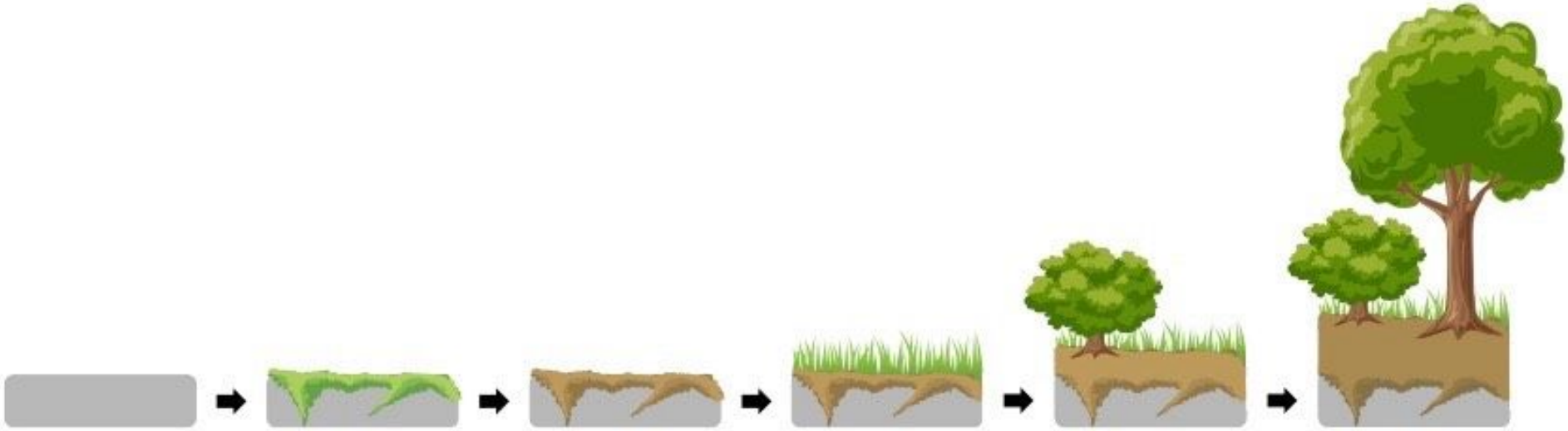
- Volcanic eruptions.
- Retreat of glaciers.
- Flooding accompanied by severe soil erosion.
- Landslides.
- Nuclear explosions.
- Oil spills.
- Abandonment of a manmade structure, such as a paved parking lot.

Secondary succession:

- Wildfire
- Floods
- Storms
- Landslides.
- Nuclear explosions.
- Minor events such as the falling of a single tree


BBC
NEWS

Primary succession



Pioneer Species

Pioneer species- are first species to populate an area during primary succession.

Examples are:

moss **lichens** **fungus** **algae**









Pioneer species will break up the rocks.



How did the pioneer species arrive?

How did the pioneer species arrive?

Wind

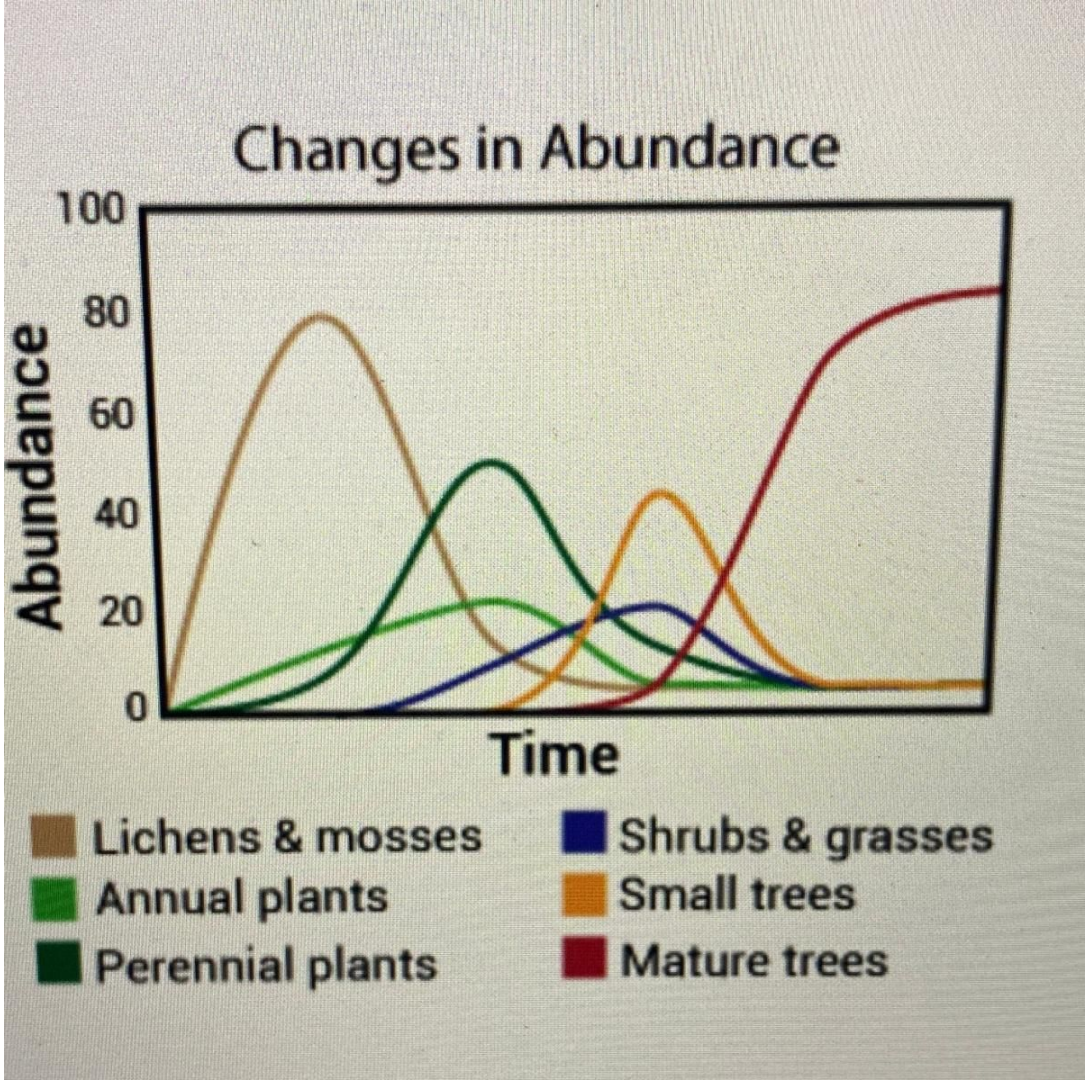
Water

Animals

General principles of succession:

- increases in size of plants
- Increase in amount of primary production
- Increase in species diversity
- Increase in complexity of food webs
- Increase in amount of nutrient cycling

Interpret this graph



Surtsey

November 14th 1963

Protected

Visited every year

Clear documentation of primary succession





Primary Succession

- Colonisation by **PIONEER SPECIES**
  - Withstand low water, no soil
- Soil produced by pioneer species
- Soil provides water and nutrients for other plants to grow
- Plants provide food and habitats for animals
- Eventually CLIMAX COMMUNITY established (balanced stable state)



What is the difference between primary and secondary succession?

Primary succession

The process begins from bare rock. **There is no soil present.**

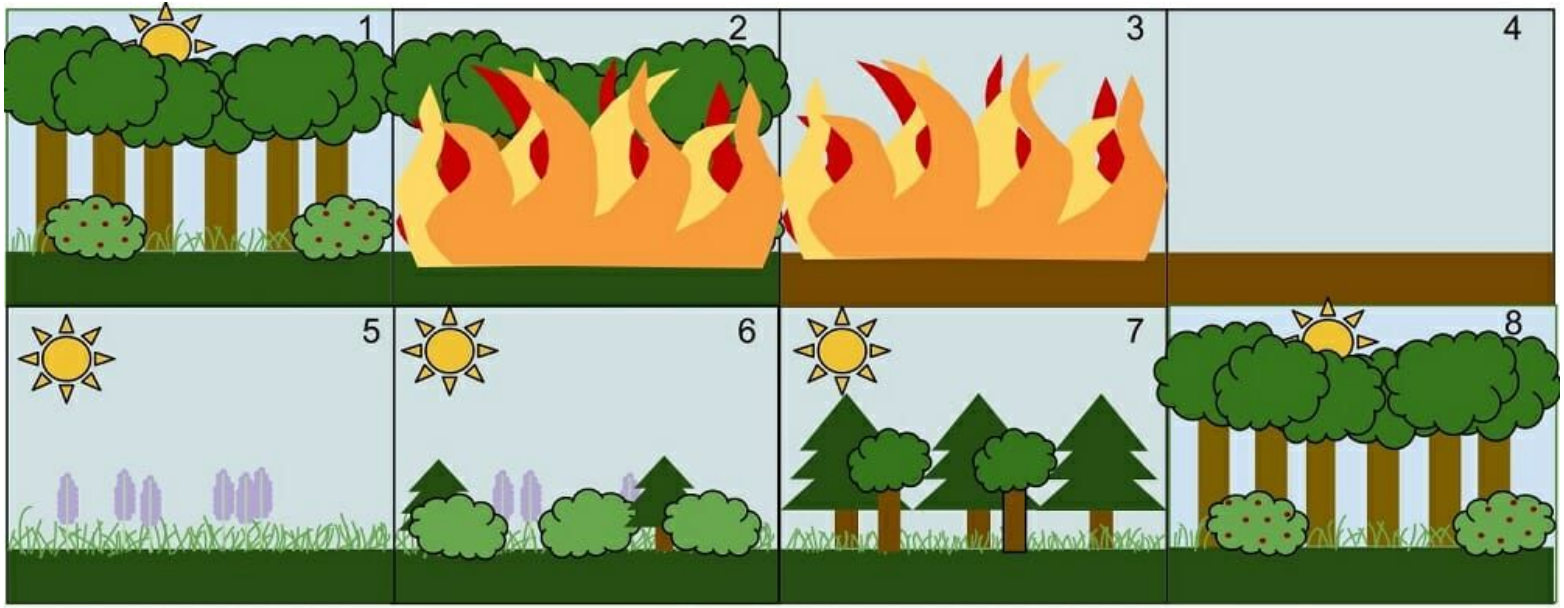
Secondary succession

The process begins with already existing soil and nutrients.

D4.2.14—Cyclical succession in ecosystems

Some ecosystems have a predictable pattern of cyclical changes.

Forest fires are a typical example of cyclical succession



D4.2.15—Climax communities and arrested succession

The stable end state of an ecosystem is called the **climax community**

Often ecosystems are never allowed to reach their climax community.

Plagioclimax communities

A **plagioclimax** community is an area or habitat in which the influences of the human race have prevented the ecosystem from developing further.

All human activity (agriculture, hunting, clearance, burning and grazing) create a plagioclimax by **arresting succession**

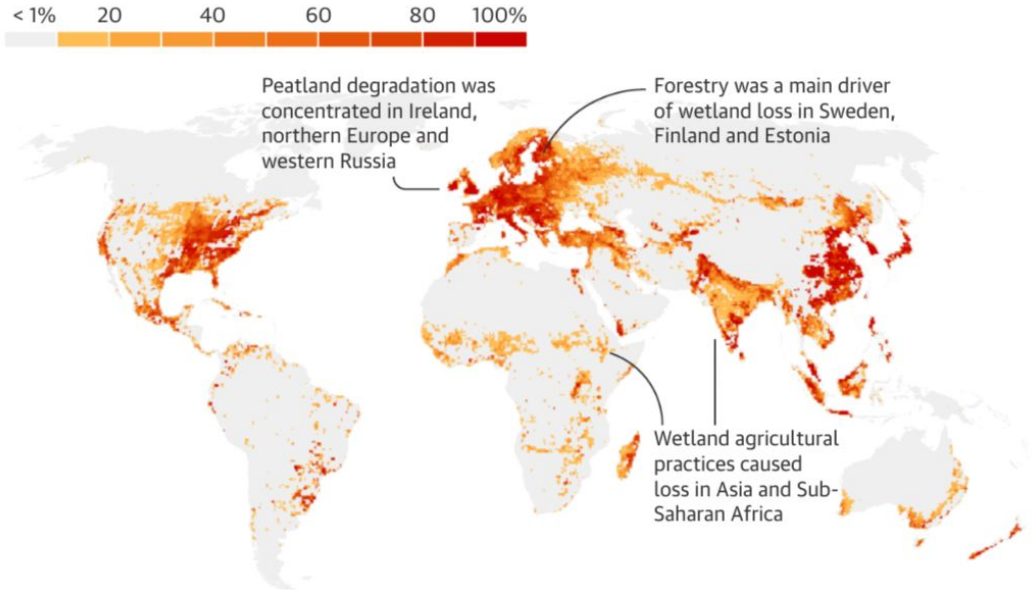
Example: Grazing by farm livestock



Example: Draining of wetlands

Wetland loss has been concentrated in the US, Europe and China

Wetland cover loss, 1700-2020



Guardian graphic. Source: Etienne Fluet-Chouinard/Nature

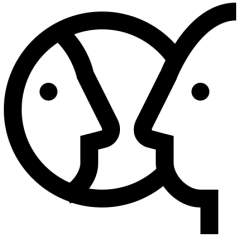
Half the wetlands in Europe lost in past 300 years, researchers calculate

Assessment shows 20% global loss since 1700 – far less than thought – but Europe, US and China bear the brunt



The Orshinski Mokh peatlands in the Tver province of Russia, where restoration efforts have been interrupted by the war in Ukraine

Can you...



Describe the process of primary succession?

Compare and contrast primary and secondary succession?

Explain a plagioclimax?